Hilbert Spaces

H.1

Solution:

□

H.2

Solution:

□

H.3

Let $u = (2, 1+i, i, 1-i)$ and $v = (1, i, -1, i)$. Calculate

$$\|u\|, \|v\|, (u \mid v), (v \mid u)$$

with respect to the ordinary scalar product in $\mathbb{C}^4$.

Solution:

$$\|u\| = \sqrt{(u \mid u)} = \sqrt{\overline{(2, 1+i, i, 1-i)} \cdot (2, 1+i, i, 1-i)} = \sqrt{4+2+1+2} = 3$$

$$\|v\| = \sqrt{(v \mid v)} = \sqrt{\overline{(1, i, -1, i)} \cdot (1, i, -1, i)} = \sqrt{1+1+1+1} = 2$$

$$(u \mid v) = \overline{(2, 1+i, i, 1-i)} \cdot (1, i, -1, i) = 2+i+1+i+i-1 = 2+3i$$

$$(v \mid u) = \overline{(1, i, -1, i)} \cdot (2, 1+i, i, 1-i) = 2-i+1-i-i-1 = 2-3i$$

□

H.4

Let

$$f(x) = 1 + ix, \quad g(x) = 2 + ix^2, \quad 0 \leq x \leq 1.$$

Calculate $(f \mid g)$ and $(g \mid f)$, where the scalar product is taken in $L_2[0,1]$.

Solution:

The scalar product is defined as

$$(u \mid v) = \int_L \overline{u} v \, \mathrm{d}x$$

This gives

$$(f \mid g) = \int_0^1 \left(\overline{1+ix}\right)(2+ix^2) \, \mathrm{d}x = \int_0^1 (2 + ix^2 - 2ix + x^3) \, \mathrm{d}x =$$

$$= \left[2x + \frac{ix^3}{3} - ix^2 + \frac{x^4}{4} \right]_0^1 = \frac{9}{4} - \frac{2}{3}i$$

and

$$(g \mid f) = \int_0^1 \left(\overline{2+ix^2}\right)(1+ix) \, \mathrm{d}x = \int_0^1 (2 + 2ix - ix^2 + x^3) \, \mathrm{d}x =$$

$$= \left[2x + ix^2 - \frac{ix^3}{3} + \frac{x^4}{4} \right]_0^1 = \frac{9}{4} + \frac{2}{3}i$$

As expected,

$$(f \mid g) = \overline{(g \mid f)}.$$

□

H.5a

Motivate why the expression

$$< f \mid g >= \int_0^1 f^2(x) g^2(x) \, \mathrm{d}x$$

is not a scalar product for real, continuous functions on the interval $[0, 1]$

Solution:

The scalar product needs to be linear, which this expression is not. For example

$$< \lambda f \mid \mu g >= \int_0^1 \lambda^2 f^2(x) \mu^2 g^2(x) \, \mathrm{d}x$$

but linearity is

$$\lambda\mu < f \mid g >= \lambda\mu \int_0^1 f^2(x) g^2(x) \, \mathrm{d}x$$

□

H.11

Verify that the vectors

$$\begin{aligned} u_1 &= (1, 1, 1, 1) & u_2 &= (1, i, -1, -i) \\ u_3 &= (1, -1, 1, -1) & u_4 &= (1, -i, -1, i) \end{aligned}$$

are pairwise orthogonal in $\mathbb{C}^4$. Determine the corresponding orthonormal base e_1, e_2, e_3, e_4 and write

$$u = (2 + i, 1 + 2i, i, 1 - 4i)$$

in this base.

Solution:

The scalar product in $\mathbb{C}^4$ is

$$(u \mid v) = \overline{u} \cdot v$$

To check that the vectors are pairwise orthogonal, we check if their scalar product is 0:

$$(u_1 \mid u_2) = 1 + i - 1 - i = 0$$
$$(u_1 \mid u_3) = 1 - 1 + 1 - 1 = 0$$
$$(u_1 \mid u_4) = 1 - i - 1 + i = 0$$

$$(u_2 \mid u_3) = 1 + i - 1 - i = 0$$
$$(u_2 \mid u_4) = 1 - 1 + 1 - 1 = 0$$

$$(u_3 \mid u_4) = 1 + i - 1 - i = 0$$

This confirms they are pairwise orthogonal. All vectors have the length 2, so the orthonormal base is

$$e_1 = \frac{u_1}{2} \qquad e_2 = \frac{u_2}{2}$$
$$e_3 = \frac{u_3}{2} \qquad e_4 = \frac{u_4}{2}$$

To find the coordinates of u in terms of e_1, e_2, e_3 and e_4, we use the projection formula. Since

$$u = \sum_{k=1}^{4} c_k e_k$$
$$\downarrow$$
$$c_k = \frac{(e_k \mid u)}{(e_k \mid e_k)} = (e_k \mid u)$$

$$c_1 = (e_1 \mid u) = \frac{1}{2}\overline{(1, 1, 1, 1)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = 2$$

$$c_2 = (e_2 \mid u) = \frac{1}{2}\overline{(1, i, -1, -i)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = 4$$

$$c_3 = (e_3 \mid u) = \frac{1}{2}\overline{(1, -1, 1, -1)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = 2i$$

$$c_4 = (e_4 \mid u) = \frac{1}{2}\overline{(1, -i, -1, i)} \cdot (2 + i, 1 + 2i, i, 1 - 4i) = -2$$

$$\downarrow$$
$$u = 2e_1 + 4e_2 + 2ie_3 - 2e_4$$

□

H.12

Let

$$f_n(x) = x^n.$$

Is $\{f_n\}_0^\infty$ pairwise orthogonal in

(a) $L_2(0,1)$

(b) $L_2(-1,1)$

Solution:

(a) No, since the scalar product in $L_2(0,1)$ is not zero:

$$\left(x^j \mid x^k\right) = \int_0^1 x^{j+k}\,\mathrm{d}x = \left[\frac{x^{j+k+1}}{j+k}\right]_0^1 = \frac{1}{j+k} \neq 0$$

(b) No,

$$\left(x^j \mid x^k\right) = \int_{-1}^1 x^{j+k}\,\mathrm{d}x = \left[\frac{x^{j+k+1}}{j+k}\right]_{-1}^1 \rightarrow \begin{cases} = 0, & \text{if } j+k+1 \text{ even} \\ \neq 0, & \text{if } j+k+1 \text{ odd} \end{cases}$$

$\square$

H.13

Let u, v in $L_2(0,1)$ be given by

$$u(x) = 1, \quad 0 \leq x \leq 1,$$
$$v(x) = x, \quad 0 \leq x \leq 1.$$

Calculate the projection of v onto u, and the projection of u onto v.

Solution:

Using the projection formula, the projection of v onto u is

$$P_{[u]}v = \frac{(u \mid v)}{(u \mid u)}u = \frac{\int_0^1 x\,\mathrm{d}x}{\int_0^1 \mathrm{d}x}1 = \frac{1}{2}$$

and the projection of u onto v is

$$P_{[v]}u = \frac{(v \mid u)}{(v \mid v)}v = \frac{\int_0^1 x \, \mathrm{d}x}{\int_0^1 x^2 \, \mathrm{d}x}x = \frac{3}{2}x$$

□

H.15

$$(u \mid v) = \int_{-\infty}^{\infty} u(x)v(x)\frac{e^{-x^2}}{\sqrt{\pi}} \, \mathrm{d}x$$

Determine the first three Hermite polynomials

Solution:

Using Grahm-Schmidt with $u_n = [1, x, x^2]$

$$\varphi_0 = 1$$

$$(\varphi_0 \mid \varphi_0) = \int_{-\infty}^{\infty} \frac{e^{-x^2}}{\sqrt{\pi}} \, \mathrm{d}x = 1$$

$$\psi_0 = \frac{\varphi_0}{(\varphi_0 \mid \varphi_0)^{\frac{1}{2}}} = 1$$

Next

$$\varphi_1 = u_1 - P_{[\varphi_0]}u_1 = u_1 - \frac{(\varphi_0 \mid u_1)}{(\varphi_0 \mid \varphi_0)}\varphi_0 = x - \frac{(1 \mid x)}{(1 \mid 1)}1 =$$

$$x - \underbrace{\int_{-\infty}^{\infty} x\frac{e^{-x^2}}{\sqrt{\pi}} \, \mathrm{d}x}_{=0 \text{ (odd function)}} = x$$

$$\psi_1 = \frac{\varphi_1}{(\varphi_1 \mid \varphi_1)^{\frac{1}{2}}} = \frac{x}{(x \mid x)^{\frac{1}{2}}}$$

$$(x \mid x) = \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}} \, \mathrm{d}x = -\frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} (x)\left(-2xe^{-x^2}\right) =$$

$$= -\frac{1}{2\sqrt{\pi}}\left(\left[xe^{-x^2}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-x^2} \, \mathrm{d}x\right) = \frac{1}{2}$$

Finally this gives

$$\psi_1 = \frac{x}{(x \mid x)^{\frac{1}{2}}} = \sqrt{2}x$$

Next

$$\varphi_2 = u_2 - P_{[\varphi_0,\varphi_1]}u_2 = x^2 - \frac{(1 \mid x^2)}{(1 \mid 1)}1 - \frac{(x \mid x^2)}{(x \mid x)}x =$$

$$= x^2 - \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x - \underbrace{\int_{-\infty}^{\infty} x^3 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x}_{=0 \text{ odd function}} = x^2 - \frac{1}{2}$$

$$\psi_2 = \frac{\varphi_2}{(\varphi_2 \mid \varphi_2)^{\frac{1}{2}}} = \left(x^2 - \frac{1}{2}\right)\left(\int_{-\infty}^{\infty}\left(x^2 - \frac{1}{2}\right)^2 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x\right)^{-\frac{1}{2}} =$$

$$= \left(x^2 - \frac{1}{2}\right)\left(\int_{-\infty}^{\infty}\left(x^4 - x^2 + \frac{1}{4}\right) \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x\right)^{-\frac{1}{2}}$$

$$\text{(i)} \int_{-\infty}^{\infty} \frac{1}{4}\frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x = \boxed{\frac{1}{4}}$$

$$\text{(ii)} \int_{-\infty}^{\infty} -x^2 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x = \boxed{-\frac{1}{2}}$$

$$\text{(iii)} \int_{-\infty}^{\infty} x^4 \frac{e^{-x^2}}{\sqrt{\pi}}\,\mathrm{d}x = -\frac{1}{2\sqrt{\pi}}\int_{-\infty}^{\infty}(x^3)\left(-2xe^{-x^2}\right)\mathrm{d}x =$$

$$= \underbrace{\left[x^2 e^{-x^2}\right]_{-\infty}^{\infty}}_{=0} - \underbrace{\int_{-\infty}^{\infty} 3x^2 e^{-x^2}\,\mathrm{d}x}_{=\frac{3\sqrt{\pi}}{2}} = \boxed{\frac{3}{4}}$$

So

$$\psi_2 = \left(x^2 - \frac{1}{2}\right)\left(\frac{1}{4} - \frac{1}{2} + \frac{3}{4}\right)^{-\frac{1}{2}} = \left(x^2 - \frac{1}{2}\right)\sqrt{2}$$

The three first orthonormal Hermite polynomials are

$$\psi_0 = \boxed{1}$$

$$\psi_1 = \boxed{\sqrt{2}x}$$

$$\psi_2 = \boxed{\sqrt{2}\left(x^2 - \frac{1}{2}\right)}$$

□

H.18

$$\int_0^\pi (x - a\sin(x) - b\sin(2x))^2 \, \mathrm{d}x$$

is minimized.

Solution:

$$v \in \mathcal{M} = [\sin(x), \sin(2x)]$$

$\sin(x)$ and $\sin(2x)$ are orthogonal and we can use the projection formula which says that the minimum is obtained when

$$v = P_{[\mathcal{M}]}u$$

$$v_1 = a\sin(x) = P_{[\sin(x)]}u = \frac{(\sin(x) \mid x)}{(\sin(x) \mid \sin(x))}\sin(x)$$

$$v_2 = b\sin(2x)$$

From this, a is calculated as

$$a = \frac{(\sin(x) \mid x)}{(\sin(x) \mid \sin(x))}$$

$$(\sin(x) \mid x) = \int_0^\pi x\sin(x)\,\mathrm{d}x = -[x\cos(x)]_0^\pi + \int_0 \pi\cos(x)\,\mathrm{d}x = \pi$$

$$(\sin(x) \mid \sin(x)) = \int_0^\pi \sin^2 x\,\mathrm{d}x = \int_0^\pi \frac{1-\cos(2x)}{2}\,\mathrm{d}x = \frac{\pi}{2}$$

$$\downarrow$$

$$\boxed{a = \frac{\pi}{\frac{\pi}{2}} = 2}$$

and b is calculated in the same way as

$$b = \frac{(\sin(2x) \mid x)}{(\sin(2x) \mid \sin(2x))}$$

$$(\sin(2x) \mid x) = \int_0^\pi x\sin(2x)\,\mathrm{d}x = -\left[\frac{x\cos(2x)}{2}\right]_0^\pi + \int_0^\pi \frac{\cos(2x)}{2}\,\mathrm{d}x = -\frac{\pi}{2}$$

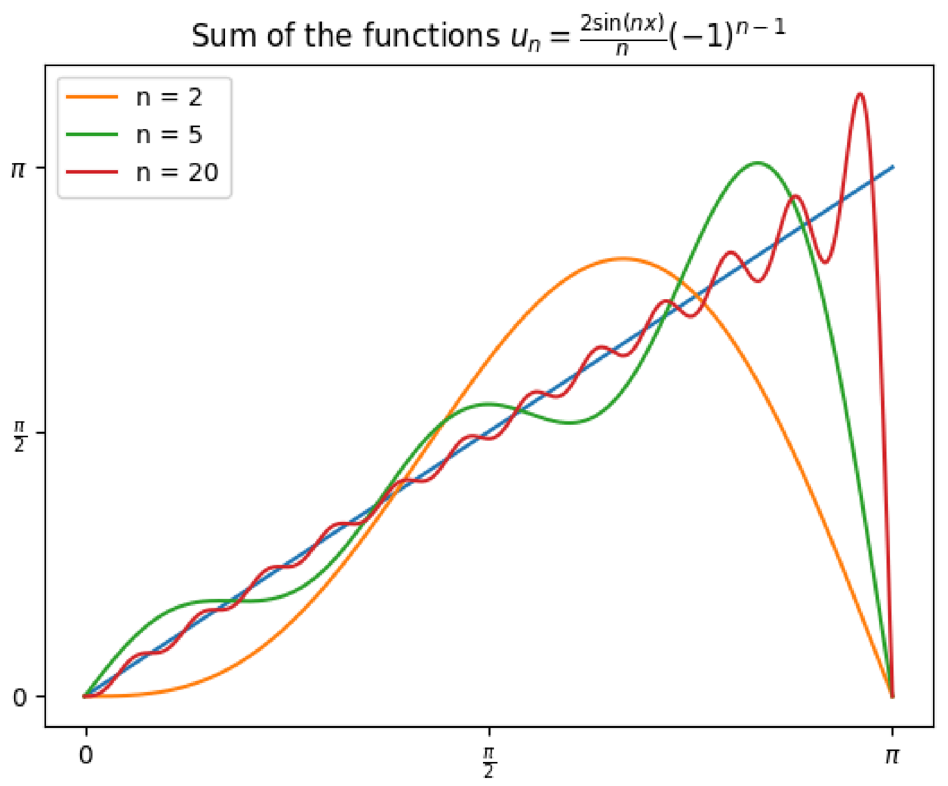
$$(\sin(2x) \mid \sin(2x)) = \int_0^\pi \sin^2(2x)\,\mathrm{d}x = \int_0^\pi \frac{1-\cos(2x)}{2}\,\mathrm{d}x = \frac{\pi}{2}$$

$$\downarrow$$

$$\boxed{b = -\frac{\frac{\pi}{2}}{\frac{\pi}{2}} = -1}$$

In general, the functions $v \in \mathcal{M} = \{\sin(nx)\}_{n=1}^{\infty}$ that best approximate $u(x) = x$ at the interval $[0, \pi]$ is given by

$$u_n = \frac{2(-1)^{n-1} \sin nx}{n}$$



□

H.19

(a) Give an orthogonal base in $\mathcal{V}$ with the scalar product

$$(u \mid v) = \int_{-1}^{1} u(x)v(x)\,\mathrm{d}x$$

(b) Let $u(x) = x^2 - x$. Determine $v \in \mathcal{V}$ so that

$$\int_{-1}^{1} |u(x) - v(x)|^2\,\mathrm{d}x$$

is minimized.

(c) Determine the lowest possible value of the integral in (b).

Solution:

(a) $\square$

H.20

Solution:

For this setup, the scalar product is defined as

$$(u \mid v) = \int_0^1 u(x)v(x)w(x)\,\mathrm{d}x = \int_0^1 u(x)v(x)x\,\mathrm{d}x$$

Find an orthogonal base

$$\varphi_0 = 1$$

$$\varphi_1 = x - P_{[\varphi_0]}x = x - \frac{(\varphi_0 \mid x)}{(\varphi_0 \mid \varphi_0)}\varphi_0 = x - 2\int_0^1 x^2\,\mathrm{d}x = x - \frac{2}{3}$$

Now $v \in \mathcal{M} = \left\{1, x - \frac{2}{3}\right\}$

Are looking for

$$\int_0^1 \left(x^4 - a\varphi_0 - b\varphi_1\right)^2 x\,\mathrm{d}x$$

From which we can calculate

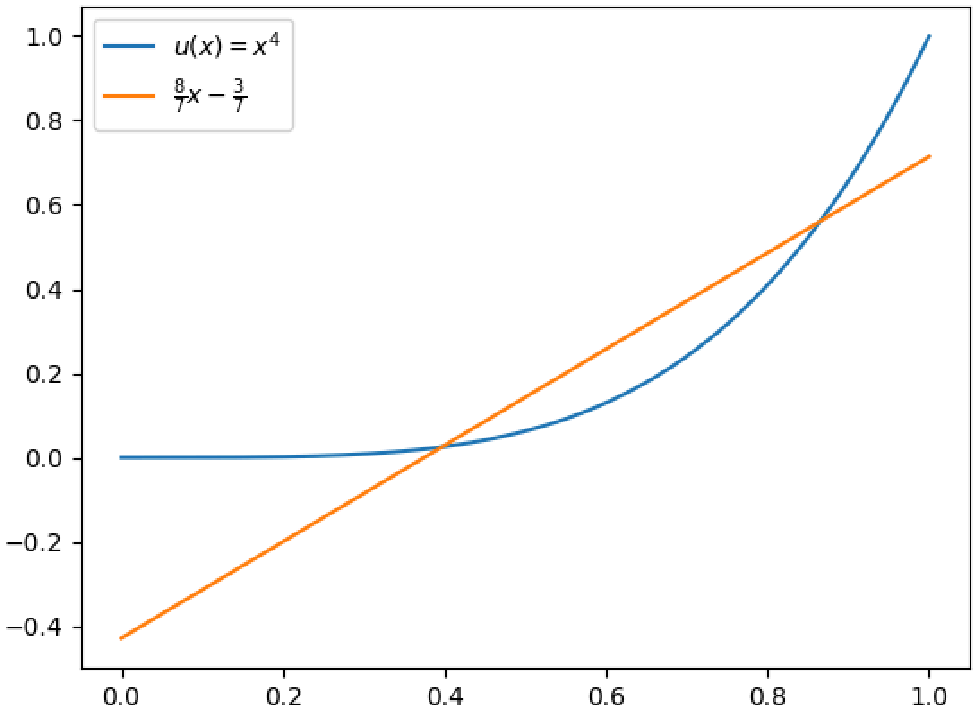
$$a = \frac{(\varphi_0 \mid x^4)}{(\varphi_0 \mid \varphi_0)} = \frac{\int_0^1 x^5\,\mathrm{d}x}{\int_0^1 x\,\mathrm{d}x} = \frac{1}{3}$$

and

$$b = \frac{(\varphi_1 \mid x^4)}{(\varphi_1 \mid \varphi_1)} = \frac{\int_0^1 \left(x^6 - \frac{2}{3}x^5\right)\mathrm{d}x}{\int_0^1 \left(x^3 - \frac{4}{3}x^2 + \frac{4}{9}x\right)} = \frac{8}{7}$$

The first degree polynomial that best approximates the function $u(x) = x^4$ in $I = [0, 1]$ is

$$\boxed{\frac{1}{3}\varphi_0 + \frac{8}{7}\varphi_1 = \frac{8}{7}x - \frac{3}{7}}$$



(b) We are now looking for a second degree polynomial which means our $\varphi \in \mathcal{M} = \{1, x, x^2\}$ φ_0 and φ_1 were already calculated in (a) so

$$\varphi_2 = u_2 - P_{[\varphi_0,\varphi_1]}u_2 = x^2 - \frac{(\varphi_0 \mid x^2)}{(\varphi_0 \mid \varphi_0)}\varphi_0 - \frac{(\varphi_1 \mid x^2)}{(\varphi_1 \mid \varphi_1)}\varphi_1$$

$$\frac{(\varphi_0 \mid x^2)}{(\varphi_0 \mid \varphi_0)}\varphi_0 = 2(\varphi_0 \mid x^2) = 2\int_0^1 x^3 \, \mathrm{d}x = \frac{1}{2}$$

$$\frac{(\varphi_1 \mid x^2)}{(\varphi_1 \mid \varphi_1)}\varphi_1 = (\varphi_1 \mid x^2)(36x - 24)$$

Now calculating

$$(\varphi_1 \mid x^2) = \int_0^1 \left(x^4 - \frac{2x^3}{3}\right) \mathrm{d}x = \frac{1}{30}$$

which gives finally

$$\varphi_2 = x^2 - \frac{6}{5}x + \frac{3}{10}$$

$$(\varphi_2 \mid \varphi_2) = \int_0^1 \left(x^2 - \frac{6x}{5} + \frac{3}{10}\right)^2 x \,\mathrm{d}x = \frac{1}{600}$$

$$\boxed{\varphi_2 = 600x^2 - 720x + 180}$$

We can now find the best approximation with

$$\int_0^1 (x^4 - a\varphi_0 - b\varphi_1 - c\varphi_2)^2 x \,\mathrm{d}x$$

$$c\varphi_2 = \frac{(\varphi_2 \mid x^4)}{(\varphi_2 \mid \varphi_2)}\varphi_2$$

$$c = \frac{(\varphi_2 \mid x^4)}{(\varphi_2 \mid \varphi_2)}$$

$$(\varphi_2 \mid x^4) = \int_0^1 (600x^2 - 720x + 180)x^4 x \,\mathrm{d}x = \frac{15}{7}$$

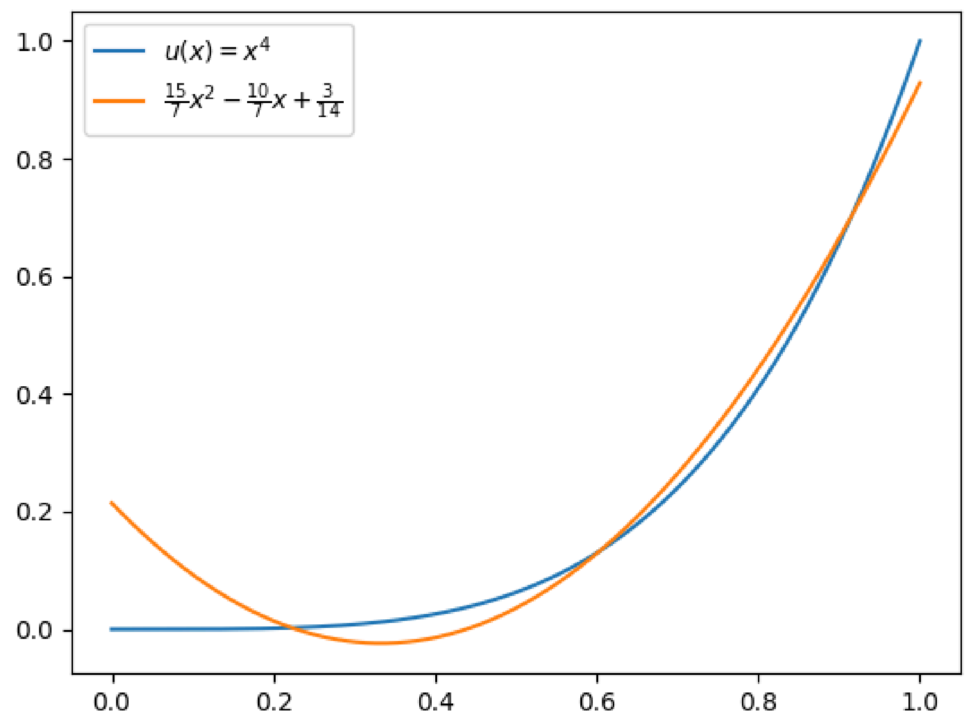
$$(\varphi_2 \mid \varphi_2) = \int_0^1 (600x^2 - 720x + 180)^2 x \,\mathrm{d}x = 600$$

$$\downarrow$$

$$c = \frac{15}{4200} = \frac{1}{280}$$

The best second degree polynomial to approximate $u(x) = x^4$ is then

$$\boxed{a\varphi_0 + b\varphi_1 + c\varphi_2 = \frac{15}{7}x^2 - \frac{10}{7}x + \frac{3}{14}}$$



□

H.27

(a) Show that p_n is orthogonal to all polynomials of degree $p < n$

(b) Show that if p is a polynomial, not identically zero, such that $(p_k \mid p) = 0$ for $k = 0, 1, 2$, then degree of $p \geq 3$.

Solution:

(a) All polynomials of degree m can be written

$$p = \sum_{k=0}^{m} c_k p_k$$

Gives

$$(p_n \mid p) = \left(p_n \mid \sum_{k=0}^{m} c_k p_k \right) = \sum_{k=0}^{m} c_k (p_n \mid p_k) = 0 \text{ for } m < n$$

since all polynomials are pairwise orthogonal

(b)

$$p = \sum_{k=0}^{m} c_k p_k$$

where

$$c_k = \frac{(p_k \mid p)}{(p_k \mid p_k)}$$

$$(p_k \mid p) = 0, \text{ for } k = 0, 1, 2 \to c_0 = c_1 = c_2 = 0$$

The restriction that $p \neq 0$ gives that $\deg p \geq 3$ □

H.30

Solution:

□

H.31

Solution:

□

H.33

$$f_n(x) = x^n, \quad x \in I = [0, 1], \quad n = 0, 1, \ldots$$

Determine $\|f_n\|_p$ for $p = 1, 2$ and ∞. Is it for any of these norms true that $f_n \to 0$ when $n \to \infty$? What is the pointwise limit function?

Solution:

$$\|f_n\|_1 = \int_0^1 |x^n| \, \mathrm{d}x = \left[\frac{x^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1} \to 0, \quad n \to \infty$$

$$\|f_n\|_2 = \left(\int_0^1 |x^n|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} = \left(\left[\frac{x^{2n+1}}{2n+1} \right]_0^1 \right)^{\frac{1}{2}} = \frac{1}{\sqrt{2n+1}} \to 0, \quad n \to \infty$$

$$\|f_n\|_\infty = \sup_{x \in [0,1]} |f_n| = \sup_{x \in [0,1]} |x^n| = 1 \to 1, \quad n \to \infty$$

□

H.36

Solution:

□

H.38

Solution:

□

H.39

A linear operator $\mathcal{A}$ on $\mathcal{H}$ is said to be unitary if it is bijective and

$$(\mathcal{A}u \mid \mathcal{A}v) = (u \mid v)$$

for all $u, v \in D_{\mathcal{A}}$. Show that every eigenvalue to a unitary operator has magnitude one.

Solution:

$$\mathcal{A}u = \lambda u, \quad \mathcal{A}v = \lambda v$$
$$\downarrow$$
$$(\mathcal{A}u \mid \mathcal{A}v) = (\lambda u \mid \lambda v) = \overline{\lambda}\lambda(u \mid v) = |\lambda|^2 \; (u \mid v) = (u \mid v)$$
$$\downarrow$$
$$|\lambda|^2 = 1$$

□

H.45

Solution:

□

H.47

Show with the help of the definition that

$$\mathcal{A} = -\frac{\mathrm{d}^2}{\mathrm{d}x^2}, \quad D_{\mathcal{A}} = \{u \in \mathcal{C}^2[0,1] \mid u'(0) = u(1) = 0\}$$

is a symmetric positive semidefinite operator on $L_2[0,1]$. Determine its eigenvalues and eigenvectors.

Solution:

Symmetric:

$$(u \mid \mathcal{A}v) = -\int_0^1 \overline{u}v'' \,\mathrm{d}x = \underbrace{-\left[\overline{u}v'\right]_0^1}_{=0} + \int_0^1 \overline{u'}v' \,\mathrm{d}x =$$

$$(*) = \int_0^1 \overline{u'}v' \,\mathrm{d}x = \underbrace{\left[\overline{u'}v\right]_0^1}_{=0} - \int_0^1 \overline{u''}v \,\mathrm{d}x = (\mathcal{A}u \mid v)$$

Positive semidefinite, start from $(*)$ and set $u = v$:

$$(*) = \int_0^1 \overline{u'}u' \,\mathrm{d}x = \int_0^1 |u'|^2 \,\mathrm{d}x \geq 0$$

Eigenvalues and eigenvectors:

$$\mathcal{A}u = \lambda u \to -u'' - \lambda u = 0$$
$$\downarrow$$
$$u'' + \lambda u \to u(x) = A\cos\left(\sqrt{\lambda}x\right) + B\sin\left(\sqrt{\lambda}x\right)$$
$$u'(0) = -\sqrt{\lambda}A\sin\left(\sqrt{\lambda}\cdot 0\right) + \sqrt{\lambda}B\cos\left(\sqrt{\lambda}\cdot 0\right) = 0 \to B = 0$$
$$u(1) = A\cos\left(\sqrt{\lambda}\cdot 1\right) = 0 \to \sqrt{\lambda} = \frac{\pi}{2} + k\pi$$
$$\downarrow$$
$$\lambda_k = \frac{\pi^2}{4} + k^2\pi^2 + k\pi^2$$
$$\varphi_k(x) = \cos\left(\sqrt{\lambda}x\right)$$

□

H.50

$$\mathcal{A} = -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\right)$$
$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(-1,1) \mid u \text{ and } u' \text{ bounded/limited}\}$$

(a) Determine a scalar product in which $\mathcal{A}$ is symmetric.

(b) Show that $\mathcal{A}$ is symmetric in the scalar product from (a), using the definition.

(c) Show that $1, x, 2x^2 - 1$ are eigenfunctions to $\mathcal{A}$ and determine the corresponding eigenvalues.

(d) Evaluate

$$\int_{-1}^{1} \frac{2x^2-1}{\sqrt{1-x^2}}\,\mathrm{d}x$$

Solution:

(a) The operator is a Sturm-Liouville operator

$$\mathcal{A}u = -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\right) = -\frac{1}{w}(pu')', \qquad w(x) = \frac{1}{\sqrt{1-x^2}}$$

and the scalar product is then

$$(u \mid v) = \int_{-1}^{1} \overline{u(x)}v(x)\frac{1}{w(x)}\,\mathrm{d}x$$

(b) Using the scalar product definition

$$(u \mid v) = \int \overline{u(x)} v(x) w(x) \, \mathrm{d}x, \qquad w(x) = \frac{1}{\sqrt{1-x^2}}$$

we get

$$(u \mid \mathcal{A} v) =$$

$$= \int_{-1}^{1} \overline{u(x)} [\mathcal{A} v(x)] \frac{1}{w(x)} \, \mathrm{d}x = \int_{-1}^{1} \overline{u(x)} \left[-\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} \left(\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} v(x) \right) \right] \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x =$$

$$= -\int_{-1}^{1} \overline{u(x)} \left(\sqrt{1-x^2} v'(x) \right)' \mathrm{d}x = -\underbrace{\left[\overline{u(x)} \sqrt{1-x^2} v'(x) \right]_{-1}^{1}}_{=0} + \int_{-1}^{1} \overline{u'(x)} \sqrt{1-x^2} v'(x) \, \mathrm{d}x =$$

$$= \int_{-1}^{1} \left(\overline{u'(x)} \sqrt{1-x^2} \right) v'(x) \, \mathrm{d}x = \underbrace{\left[\left(\overline{u'(x)} \sqrt{1-x^2} \right) v(x) \right]_{-1}^{1}}_{=0} - \int_{-1}^{1} \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \, \mathrm{d}x =$$

$$= \int_{-1}^{1} -\left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \, \mathrm{d}x = \int_{-1}^{1} -\sqrt{1-x^2} \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x =$$

$$= \int_{-1}^{1} \left[-\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} \left(\sqrt{1-x^2} \frac{\mathrm{d}}{\mathrm{d}x} \overline{u(x)} \right) \right] v(x) \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x =$$

$$= \int_{-1}^{1} \left[\mathcal{A} \overline{u(x)} \right] v(x) \frac{1}{w(x)} \, \mathrm{d}x = \boxed{(\mathcal{A} u \mid v)}$$

(c)

$$\mathcal{A}u = \lambda u$$
$$u(x) = 1$$
$$\downarrow$$
$$\mathcal{A} \cdot 1 = \lambda_0 \cdot 1 \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x} \cdot 1\right) = \lambda_0 \cdot 1 \to 0 = \lambda_0 \cdot 1 \to \boxed{\lambda_0 = 0}$$

$$u(x) = x$$
$$\downarrow$$
$$\mathcal{A}x = \lambda_1 x \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x} x\right) = \lambda_1 x \to -\sqrt{1-x^2}\frac{-2x}{2\sqrt{1-x^2}} = \lambda_1 x \to$$
$$\to x = \lambda_1 x \to \boxed{\lambda_1 = 1}$$

$$u(x) = 2x^2 - 1$$
$$\downarrow$$
$$\mathcal{A}(2x^2-1) = \lambda_2(2x^2-1) \to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}(2x^2-1)\right) = \lambda_2(2x^2-1) \to$$
$$\to -\sqrt{1-x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(4x\sqrt{1-x^2}\right) = -\sqrt{1-x^2}\left(4\sqrt{1-x^2} - \frac{8x^2}{2\sqrt{1-x^2}}\right) = -4 + 4x^2 + 4x^2 =$$
$$= 8x^2 - 4 = \lambda_2(2x^2-1) \to 4(2x^2-1) = \lambda_2(2x^2-1) \to \boxed{\lambda_2 = 4}$$

(d) The integral can be rewritten with the scalar product in (a) as

$$\int_{-1}^{1} \frac{2x^2-1}{\sqrt{1-x^2}}\,\mathrm{d}x = (1 \mid 2x^2-1), \qquad w(x) = \frac{q}{\sqrt{1-x^2}}$$

Since the functions 1 and $2x^2-1$ are eigenfunctions, the scalar product between them are pairwise orthogonal and therefore $= 0$.

□

H.51

Solution:

□

H.52

Solution:

□

Operator

$$\mathcal{A} = -\ \texttt{laplace} = -\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$$

Solution:

$$(u \mid \mathcal{A}v) = \int_0^a \int_0^b \overline{u}(-v''_{xx} - v''_{yy})\,\mathrm{d}x\,\mathrm{d}y =$$

$$(1) = -\int_0^b \underbrace{\left(\int_0^a \overline{u}v''_{xx}\,\mathrm{d}x\right)}_{\text{(i-x)}}\mathrm{d}y - \int_0^a \underbrace{\left(\int_0^b \overline{u}v''_{yy}\,\mathrm{d}y\right)}_{\text{(i-y)}}\mathrm{d}x$$

$$\text{(i-x)} = \underbrace{\left[\overline{u}v'_x\right]_0^a}_{=0} - \int_0^a \overline{u}'_x v'_x\,\mathrm{d}x$$

$$\text{(i-y)} = \underbrace{\left[\overline{u}v'_y\right]_0^b}_{=0} - \int_0^b \overline{u}'_y v'_y\,\mathrm{d}y$$

$$\downarrow$$

$$(1) = \int_0^b \underbrace{\left(\int_0^a \overline{u}'_x v'_x\,\mathrm{d}x\right)}_{\text{(ii-x)}}\mathrm{d}y + \int_0^a \underbrace{\left(\int_0^b \overline{u}'_y v'_y\,\mathrm{d}y\right)}_{\text{(ii-y)}}\mathrm{d}x = (***)$$

$$\text{(ii-x)} = \underbrace{\left[\overline{u}'_x v'_x\right]_0^a}_{=0} - \int_0^a \overline{u}''_{xx} v\,\mathrm{d}x$$

$$\text{(ii-y)} = \underbrace{\left[\overline{u}'_y v'_y\right]_0^b}_{=0} - \int_0^b \overline{u}''_{yy} v\,\mathrm{d}y$$

$$\downarrow$$

$$(1) = -\int_0^b \int_0^a \overline{u}''_{xx} v\,\mathrm{d}x\,\mathrm{d}y - \int_0^a \int_0^b \overline{u}''_{yy} v\,\mathrm{d}y\,\mathrm{d}x = \int_0^a \int_0^b (-\overline{u}''_{xx} - \overline{u}''_{yy})v\,\mathrm{d}x\,\mathrm{d}y =$$

$$= \boxed{(\mathcal{A}u \mid v)}$$

To show positive semidefinite, use the form described on (***) and $u = v$:

$$(***) = \int_0^a \int_0^b (\overline{u}'_x u'_x + \overline{u}'_y u'_y)\,\mathrm{d}x\,\mathrm{d}y = \int_0^a \int_0^b \left(|u'_x|^2 + |u'_y|^2\right)\mathrm{d}x\,\mathrm{d}y \geq 0$$

To find the eigenvalues associated with the operator, use separation of variables

$$u(x, y) = X(x)Y(y)$$

$$\mathcal{A}u = \lambda u \to -X''Y - XY'' = \lambda XY \to -\frac{X''}{X} - \frac{Y''}{Y} = \lambda$$

$$\downarrow$$

$$\begin{cases} \frac{X''}{X} = -\gamma_x^2, & X(0) = X(a) = 0 \\ \frac{Y''}{Y} = -\gamma_y^2, & Y'(0) = Y'(b) = 0 \end{cases}$$

Starting with $X(x)$:

$$X'' = -\gamma_x^2 X \to X(x) = e^{i\gamma_x x} = A\cos(\gamma_x x) + B\sin(\gamma_x x)$$
$$X(0) = A = 0 \to A = 0$$
$$X(a) = B\sin(\gamma_x a) = 0 \to \gamma_x = \frac{\pi k_x}{a},\ k_x = 1, 2, 3...$$
$$\downarrow$$
$$X(x) = B\sin\left(\frac{\pi k_x x}{a}\right) \quad \square$$

Now $Y(y)$:

$$Y'' = -\gamma_y^2 Y \to Y(y) = e^{i\gamma_y y} = A\cos(\gamma_y y) + B\sin(\gamma_y y)$$
$$Y'(x) = -\gamma_y A\sin(\gamma_y y) + \gamma_y B\cos(\gamma_y y)$$
$$Y'(0) = \gamma_y B = 0 \to B = 0$$
$$Y'(b) = -\gamma_y A\sin(\gamma_y b) = 0 \to \gamma_y = \frac{\pi k_y}{b},\ k_y = 0, 1, 2...$$
$$\downarrow$$
$$Y(y) = A\cos\left(\frac{\pi k_y y}{b}\right) \quad \square$$

$\square$

H.8 Operators in Hilbert Spaces

Linear avbildning

$$\mathcal{A} : \mathcal{H}1 \to \mathcal{H}2$$

where $\mathcal{H}1$ and $\mathcal{H}2$ are two pre-Hilbert Spaces. Linearity:

$$\mathcal{A}(\lambda u + \mu v) = \lambda\mathcal{A}u + \mu\mathcal{A}v$$

is termed a **linear operator**.

Example H.19

Let I be a compact interval on $\mathbb{R}$, and let $K(x, y)$ be continuous for $x, y \in I$.

$$v(x) = \int_I K(x, y)u(y)\,\mathrm{d}y$$

This defines an **integral operator**

$$\mathcal{K} : L_2(I) \to L_2(I)$$

The function $K(x, y)$ is termed the operators **kernel**